

LATTICE-BASED MESOSCALE SIMULATIONS OF CELL MEMBRANE ADHESION AND DISTRIBUTION OF LIPID RAFTS

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ABSTRACT

Cell membrane is an interface that separates the intracellular fluids and organelles from the extracellular environment [1]. This interface serves as a platform to many relevant biological processes such as tissue formation, cell-cell communication, nutrient uptake and drug efflux. The cell membrane is composed of diverse macromolecules-primarily lipids and proteins which govern various biological processes. The interaction between two apposing cells is mediated through a process called membrane adhesion which arises from the specific binding of membrane-anchored receptor proteins to their ligands present in the apposing membrane. This physical contact of apposing cells can cause significant changes in the spatial distribution of membrane proteins and lipids affecting the dynamic rearrangement of cell membranes. Here, we employ a statistical-mechanical model and perform Monte Carlo simulations to study a system of adhered membranes in which the receptor and ligand proteins exhibit affinity for association with so-called lipid rafts, which are fluctuating nanoscale molecular clusters enriched in sphingolipid and cholesterol. We focus on equilibrium properties of the adhered membranes in the mixed phase, where both the membrane-anchored proteins and lipid rafts are distributed more-or-less uniformly within the membranes. Our simulation results show that lateral attraction between lipid rafts enhances the receptor-ligand binding, affecting the adhesion of the membranes. On the other hand, the receptor-ligand binding causes lipid rafts to be distributed less uniformly within the membranes and, simultaneously, leads to an increased co-localization of the membrane-anchored proteins with lipid rafts. We quantify and discuss all these effects, providing a detailed picture of the complex interplay between the

adhesion of the membranes and the lateral distribution of the membrane-anchored proteins and lipid rafts [2]. Our results broaden the understanding of the physical mechanisms that determine the supra-molecular organization of lipid rafts and membrane receptors in cell membranes. This understanding may help to elucidate how lipid rafts function in biological processes such as cell signaling and immune responses.

REFERENCES

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